CSA0405-OPERATING SYSTEM

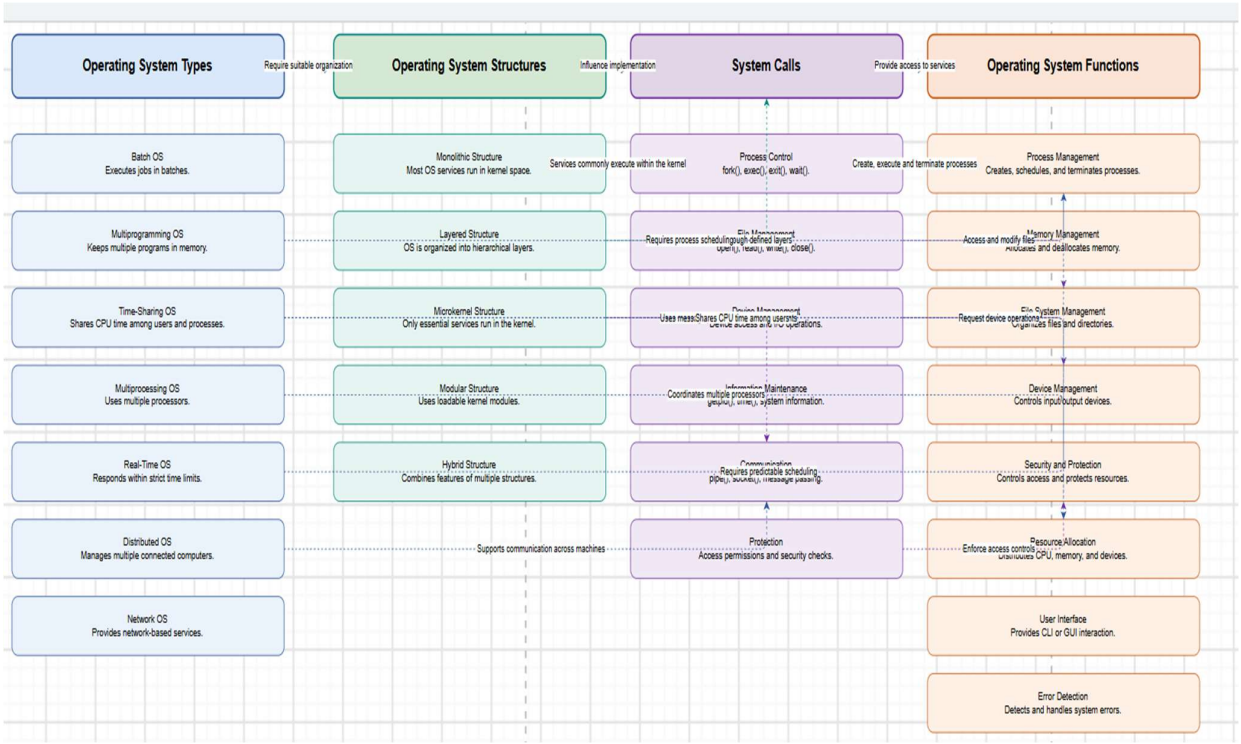
CO1-AT1

TECHNICAL PRESENTATION

OPERATING SYSTEM:TYPES,STRUCTURES,SYSTEMCALLS,FUNCTIONS

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Operating System Types
Require suitable organization
Operating System Structures
Influence implementation
System Calls
Provide access to services
Operating System Functions
Batch OS
Executes jobs in batches.
Monolithic Structure
Most OS services run in kernel space.
Services commonly execute within the kernel
Process Control
fork(), exec(), exit(), wait().
Create, execute and terminate processes
Process Management
Creates, schedules, and terminates processes.
Multiprogramming OS
Keeps multiple programs in memory.
Layered Structure
OS is organized into hierarchical layers.
Requires process scheduling
ough defined layers.
open(), read(), write(), close().
Access and modify files
Memory Management
Allocates and deallocates memory.
Time-Sharing OS
Shares CPU time among users and processes.
Microkernel Structure
Only essential services run in the kernel.
Uses mess
Shares CPU time among users.
Request device operations
File System Management
Organizes files and directories.
Multiprocessing OS
Uses multiple processors.
Modular Structure
Uses loadable kernel modules.
Coordinates multiple processors
Device Management
Controls input/output devices.
Real-Time OS
Responds within strict time limits.
Hybrid Structure
Combines features of multiple structures.
Requires predictable scheduling
Security and Protection
Controls access and protects resources.
Distributed OS
Manages multiple connected computers.
Supports communication across machines
Protection
Access permissions and security checks.
Enforce access controls
Resource Allocation
Distributes CPU, memory, and devices.
Network OS
Provides network-based services.
User Interface
Provides CLI or GUI interaction.
Error Detection
Detects and handles system errors.